

# Supplementary Material: Operator-Specific Forgetting in Matrix-Valued Associative Memory

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## 1 A. Reproducibility manifest

Reference commit: cec4ffe. Tag: lab-radioactivo-v1. Repository: <https://github.com/01javierescobar/operator-specific-forgetting>. Archived DOI: 10.5281/zenodo.22051928.

The repository contains implementations, execution runners, evaluation probes, audit logs, model checkpoints, frozen claims, and execution transcripts for experimental milestones O01–O05. The primary trained-model gate is O05. Verification tooling:

- `paper/verify_numbers.py`: Recomputes all quantitative values and consistency tolerances directly from frozen JSON artifacts.
- `paper/make_figures.py`: Regenerates all publication figures (PDF and PNG at 300 DPI) from the same frozen artifacts.

## 2 B. Notation and memory construction

$D$  denotes the key and value dimension ( $D \in \{64, 128\}$ ).  $n$  is the number of stored key–value pairs.  $c = (n - 1)/D$  is the associative crosstalk load excluding the target item.  $\mathbf{w}_i \in \mathbb{C}^D$  are fixed codebook keys with unit-modulus entries ( $|\mathbf{w}_{i,k}| = 1$ , squared Euclidean norm  $\|\mathbf{w}_i\|_2^2 = D$ ).  $\mathbf{v}_i \in \mathbb{R}^D$  are real-valued target vectors. The memory state  $\mathbf{M} \in \mathbb{C}^{D \times D}$  stores associations via matrix superposition:

$$\mathbf{M} = \sum_{i=1}^n \mathbf{w}_i \mathbf{v}_i^\top. \quad (1)$$

The matched-filter query read using key  $\mathbf{w}_j$  is computed as  $\hat{\mathbf{v}}_j = \mathbf{w}_j^H \mathbf{M} / D$ . The pooled residual metric is defined as

$$L = \frac{\sum_i \|\mathbf{r}_i\|^2}{\sum_i \|\mathbf{v}_i\|^2}, \quad (2)$$

where  $\mathbf{r}_i$  is the post-erase query readout residual.

## 3 C. Formal operator definitions

### 3.1 C.1 Key drift reread erase

Memory items are stored unperturbed. At erase time, local oscillator drift alters the erase key to  $\tilde{\mathbf{w}}_j = \mathbf{w}_j e^{i\delta}$ . The erase operator rereads using  $\tilde{\mathbf{w}}_j$  and subtracts the outer product:

$$\hat{\mathbf{v}}_j = \frac{\tilde{\mathbf{w}}_j^H \mathbf{M}}{D}, \quad \mathbf{M}' = \mathbf{M} - \tilde{\mathbf{w}}_j \hat{\mathbf{v}}_j. \quad (3)$$

Because both the projection estimate and the subtractive unbinding use  $\tilde{\mathbf{w}}_j$ , the operation represents an exact orthogonal projection onto the subspace orthogonal to  $\tilde{\mathbf{w}}_j$ .

### 3.2 C.2 State drift reread erase

We use state drift to denote a write-time phase offset attached to the target association, followed by a mismatch between the stored phase coordinate used for rereading and the reference coordinate used for unbinding. Specifically, the target association is stored as  $(e^{i\delta}\mathbf{w}_j)\mathbf{v}_j^\top$ . At erase time, the erase operator rereads using the item’s stored key coordinate  $\tilde{\mathbf{w}}_j = e^{i\delta}\mathbf{w}_j$  to recover  $\hat{\mathbf{v}}_j = \tilde{\mathbf{w}}_j^H \mathbf{M}/D$ , but subtracts using the current reference clock key  $\mathbf{w}_j$ :

$$\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \hat{\mathbf{v}}_j. \quad (4)$$

### 3.3 C.3 Real-truncated readout channel

When evaluating the real channel, real truncation is applied to the reread estimate before subtraction:  $\hat{\mathbf{v}}_j \leftarrow \text{Re}(\hat{\mathbf{v}}_j)$ . At query time, the readout residual is evaluated as  $\mathbf{r}_j = \text{Re}(\mathbf{w}_j^H \mathbf{M}'/D)$ .

### 3.4 C.4 Representational erase control

The erased value is re-presented from an external target rather than reread from the state:  $\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \mathbf{v}_j^\top$ . This serves as a control condition separating operator-level self-estimation from state residual dynamics.

This control was executed at  $D = 64$  in an early milestone (three seeds  $\times$  six drift magnitudes, both arms). Measured pooled residuals track the parameter-free closed-form predictions within the consistency tolerance on each arm’s own readout channel: a maximum absolute deviation of 0.020 (complex) and 0.017 (Re) over seeds and drift grid. Because these predictions do not depend on how the perturbation is injected, the control isolates self-estimation error from intrinsic post-erase residual structure.

### 3.5 C.5 Normalized delta control

The corrective delta-rule update modifies state  $\mathbf{S} \in \mathbb{R}^{D \times D}$  via

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \beta \mathbf{k}_t (\mathbf{v}_t^\top - \mathbf{k}_t^\top \mathbf{S}_{t-1}), \quad (5)$$

where  $\mathbf{k}_t$  and  $\mathbf{v}_t$  are column vectors and the outer product of  $\mathbf{k}_t$  with the row residual  $(\mathbf{v}_t^\top - \mathbf{k}_t^\top \mathbf{S}_{t-1})$  is taken; for complex keys and states, transposes are replaced by conjugate transposes. In the normalized-key delta control arm, keys are normalized ( $\|\mathbf{k}_t\|_2^2 = 1$ ). For  $0 < \beta < 2$ , this keeps the directional feedback multiplier  $|1 - \beta| < 1$  strictly inside the stable unit disk and maintains a bounded state norm.

## 4 D. Closed-form law derivations

Throughout this section, key entries are drawn i.i.d. with independent uniform phases over  $[0, 2\pi)$ . This yields  $\mathbb{E}[\mathbf{w}_i^H \mathbf{w}_k] = 0$  and  $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k|^2] = D$  for  $i \neq k$ , vanishing mixed cross-moments ( $\mathbb{E}[\alpha_{jk} \alpha_{jl}^*] = 0$  for  $k \neq l$ , where  $\alpha_{jk} = \mathbf{w}_j^H \mathbf{w}_k/D$ ), and circular crosstalk phasors ( $\mathbb{E}[z^2] = 0$  and  $\mathbb{E}[(\text{Re } z)^2] = \mathbb{E}[|z|^2]/2$  for the Re-channel statements). Target vectors are deterministic and independent of the keys.

### 4.1 D.1 Distributed crosstalk scaling

For independent random unit-modulus keys  $\mathbf{w}_i, \mathbf{w}_k \in \mathbb{C}^D$ , the inner products satisfy  $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k|^2] = D$  for  $i \neq k$ . After normalization by  $D$ , each interfering association contributes expected key-coefficient energy  $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k/D|^2] = 1/D$  per unit squared norm of its target vector (absolute energy  $\|\mathbf{v}_k\|^2/D$ ). Summing over  $n - 1$  interfering associations yields the crosstalk load  $c = (n - 1)/D$ , consistent with distributed associative memory capacity [1–3].

## 4.2 D.2 Key drift under complex channel

The operator  $\mathbf{P}_{\tilde{\mathbf{w}}_j}^\perp = \mathbf{I} - \tilde{\mathbf{w}}_j \tilde{\mathbf{w}}_j^H / D$  is the orthogonal projection onto the  $(D - 1)$ -dimensional subspace orthogonal to  $\tilde{\mathbf{w}}_j$  (the rank-one object is  $\tilde{\mathbf{w}}_j \tilde{\mathbf{w}}_j^H / D$ ). When querying with reference key  $\mathbf{w}_j = \tilde{\mathbf{w}}_j e^{-i\delta}$ , the target component in  $\mathbf{M}'$  along direction  $\tilde{\mathbf{w}}_j$  is identically zero. Therefore,  $L_{\text{key, complex}}(\delta) = 0$  in the algebraic model.

## 4.3 D.3 State drift under complex channel

With target item stored as  $(e^{i\delta} \mathbf{w}_j) \mathbf{v}_j^\top$  and unbinding performed with reference key  $\mathbf{w}_j$ , the reread estimate is:

$$\hat{\mathbf{v}}_j = \frac{\tilde{\mathbf{w}}_j^H \mathbf{M}}{D} = \mathbf{v}_j^\top + \frac{e^{-i\delta}}{D} \sum_{k \neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \quad (6)$$

The target phase cancels because the estimate is read with the item's own stored coordinate  $\tilde{\mathbf{w}}_j$ , while the crosstalk terms retain the  $e^{-i\delta}$  factor. Subtracting  $\mathbf{w}_j \hat{\mathbf{v}}_j$  yields the post-erase state:

$$\mathbf{M}' = (e^{i\delta} - 1) \mathbf{w}_j \mathbf{v}_j^\top + \sum_{k \neq j} \mathbf{w}_k \mathbf{v}_k^\top - \mathbf{w}_j \frac{e^{-i\delta}}{D} \sum_{k \neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \quad (7)$$

Reading with reference key  $\mathbf{w}_j$  yields target residual  $(e^{i\delta} - 1) \mathbf{v}_j^\top$  and interference residual

$$\mathbf{r}_{\text{cross}} = (1 - e^{-i\delta}) \frac{1}{D} \sum_{k \neq j} (\mathbf{w}_j^H \mathbf{w}_k) \mathbf{v}_k^\top. \quad (8)$$

Squared residual norms are summed over stored items before taking expectations (matching the definition of the pooled estimator). Substituting  $\mathbb{E}[|\mathbf{w}_j^H \mathbf{w}_k|^2] = D$  for  $k \neq j$  and  $\sum_j \sum_{k \neq j} \|\mathbf{v}_k\|^2 = (n - 1) \sum_k \|\mathbf{v}_k\|^2$  gives

$$\frac{\sum_j \mathbb{E}[\|\mathbf{r}_j\|^2]}{\sum_j \|\mathbf{v}_j\|^2} = |e^{i\delta} - 1|^2 + |1 - e^{-i\delta}|^2 \frac{n - 1}{D} = 2(1 - \cos \delta)(1 + c), \quad (9)$$

with no equal-norm assumption on the target vectors. (Per-item expectations would additionally require equal target norms and are not invoked.)

## 4.4 D.4 Real-truncated channel derivations

Applying the real projection operator  $\text{Re}(\cdot)$  to both the reread estimate ( $\hat{\mathbf{v}}_j \leftarrow \text{Re}(\hat{\mathbf{v}}_j)$ ) and the final query readout ( $\mathbf{r}_j = \text{Re}(\mathbf{w}_j^H \mathbf{M}' / D)$ ) projects out the imaginary component of the phase differences.

- **Key drift (Re channel):** Under key drift, the target contribution to the real query residual is  $\sin^2 \delta \cdot \mathbf{v}_j^\top$ , yielding target energy  $\sin^4 \delta \|\mathbf{v}_j\|^2$ . The interfering crosstalk from  $n - 1$  unperturbed associations integrates over independent random phases to pooled relative energy  $\frac{c}{4}(1 - \cos 2\delta)$ . Cross-terms between target and interference vanish in expectation due to zero-mean key correlations ( $\mathbb{E}[\mathbf{w}_j^H \mathbf{w}_k] = 0$  for  $k \neq j$ ). Summing gives:

$$L_{\text{key, Re}}(\delta) = \sin^4 \delta + \frac{c}{4}(1 - \cos 2\delta). \quad (10)$$

- **State drift (Re channel):** The target association written with phase offset  $e^{i\delta}$  yields an unbinding target residual  $(\cos \delta - 1) \mathbf{v}_j^\top$ , with energy  $(1 - \cos \delta)^2 \|\mathbf{v}_j\|^2$ . The unbinding subtracts the real part of the reread crosstalk, leaving pooled interference energy  $c(1 - \cos \delta)$ . Independent key phases ensure vanishing cross-terms in expectation, yielding:

$$L_{\text{state, Re}}(\delta) = (1 - \cos \delta)^2 + c(1 - \cos \delta). \quad (11)$$

## 5 E. Write stability analysis

Along direction  $\mathbf{k}_t$ , projecting the state error yields the first-order scalar recursion

$$s_t = (1 - \beta \|\mathbf{k}_t\|^2) s_{t-1} + \beta q_t, \quad (12)$$

where  $q_t$  represents crosstalk from other stored patterns. The homogeneous dynamics are stable if and only if  $|1 - \beta \|\mathbf{k}_t\|^2| < 1$ , which requires  $0 < \beta \|\mathbf{k}_t\|^2 < 2$ .

With  $\beta = 1$  and unnormalized embedding initialization ( $\|\mathbf{k}_t\|^2 \approx 8$ ), the multiplier  $|1 - 8| = 7$  compounds along each revisited key direction—the task draws from a finite key vocabulary that recurs across updates—inducing exponential growth of the state norm ( $\|\mathbf{S}\| \approx 10^{11}$ – $10^{12}$  within early epochs). Key normalization ensures  $\|\mathbf{k}_t\|_2^2 = 1$ ; for  $0 < \beta < 2$ , this yields  $|1 - \beta| < 1$  and keeps the directional dynamics inside the unit disk.

## 6 F. Experimental protocol (O05 milestone)

- **Dimensions:** Primary scale  $D = 128$  ( $n = 40$  pairs, predicted load  $c = 39/128 \approx 0.3047$ , five seeds) compared against  $D = 64$  ( $n = 20$  pairs, predicted load  $c = 19/64 \approx 0.2969$ , three seeds), achieving approximately matched load ( $\approx 2.6\%$  difference).
- **Vocabulary:** Discrete alphabet of 105 tokens (48 key tokens, 48 value tokens, special marker tokens). The token vocabulary is used strictly to construct controlled key–value associative sequences; the evaluation is not intended as a natural-language modeling benchmark.
- **Arms:** `wave_complex`, `wave_re`, `normalized_delta`.
- **Training:** AdamW (learning rate  $10^{-3}$ , weight decay 0.01, gradient-norm clip 1.0). Wave arms: 60 epochs over 600 training examples; `normalized_delta`: 150 epochs over 1200 examples. Model dimension 128, two layers. Exact operator definitions are in `prototypes/` in the repository.
- **Seeds and codebook:** The unit-modulus key codebook is fixed across all seeds (generation seed 0). The random seed controls parameter initialization and the dataset draws (train/validation pairs and probe events); seed-level variability therefore reflects optimization and data-sampling variability, not resampling of the random-key ensemble over which the closed-form expectations are taken.
- **Events and Units:** 320 forgetting events per arm and seed (exceeding pre-specified minimum of 300). For all inferential statistical tests, the random seed was treated as the independent experimental unit.
- **Drift grid:**  $\delta \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$ .
- **Ridge probe:** Linear Ridge regression ( $\alpha = 0.1$ , within-model split of 75% train [ $n = 240$ ] / 25% test [ $n = 80$ ]) mapping readout residuals  $\mathbf{r}_j$  to continuous targets  $\mathbf{v}_j \in \mathbb{R}^D$ . Continuous predictions were decoded to the discrete value vocabulary by nearest cosine distance, with exact match recorded when the recovered token matched the true target token; chance level is  $1/48 \approx 2.1\%$  over the 48-value vocabulary.
- **Selectivity:**  $\text{Selectivity} = \text{EM}_{\text{alive}} - \text{EM}_{\text{erased}}$ .
- **Equivalence test:** Two one-sided  $t$ -tests (TOST at  $\alpha = 0.05$ , two-sided 90% CI) are computed on retrieval selectivity between `wave_complex` and `normalized_delta` with pre-specified margin  $\varepsilon = 0.02$ . Differences are computed per seed (paired across arms) and tested against the margin with a Student- $t$  interval ( $\text{df} = 4$ ).

## 7 G. Numerical results summary

At  $D = 128$ , all 15 seed–arm combinations reached  $\text{EM} = 1.000$ ; at  $D = 64$ , the nine seed–arm combinations trained to saturation ( $\text{EM} \geq 0.99$  in every recorded run). The four closed-form laws predicted empirical pooled residuals with maximum absolute deviation of 0.0049 at  $D = 128$  and 0.0094 at  $D = 64$ , remaining below the 0.02 consistency tolerance. Dimension-matched load agreement for the three non-trivial cells showed maximum cross-scale difference 0.0064 under the stored pairing convention (below 0.011 under every aggregation convention examined), of which 0.0019 (state/complex,  $\delta = 0.5$ ) is attributable to the residual load mismatch alone.

Per-seed dispersion of the pooled deviations is reported in Table 1: the stored maxima are not driven by a single outlier seed.

Table 1: Maximum absolute deviation  $|\text{measured} - \text{predicted}|$  per seed under the pooled estimator (recomputed from frozen artifacts). For  $D = 64$ , each entry is the maximum over the two stored pooled values of that seed; “max” is the value quoted in Table 1 of the main text.

| $D$ | cell/channel  | $s_1$       | $s_2$       | $s_3$       | $s_4$       | $s_5$       | max         |
|-----|---------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 128 | key/complex   | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ |
| 128 | state/complex | 0.0031      | 0.0004      | 0.0033      | 0.0049      | 0.0006      | 0.0049      |
| 128 | key/Re        | 0.0006      | 0.0007      | 0.0018      | 0.0018      | 0.0013      | 0.0018      |
| 128 | state/Re      | 0.0018      | 0.0006      | 0.0015      | 0.0011      | 0.0002      | 0.0018      |
| 64  | key/complex   | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ | –           | –           | $< 10^{-4}$ |
| 64  | state/complex | 0.0094      | 0.0077      | 0.0086      | –           | –           | 0.0094      |
| 64  | key/Re        | 0.0009      | 0.0016      | 0.0007      | –           | –           | 0.0016      |
| 64  | state/Re      | 0.0015      | 0.0003      | 0.0015      | –           | –           | 0.0015      |

Because all closed-form laws are expectations over the random-key ensemble while every trained experiment shares a single fixed codebook (Supp. F), the consistency tolerance was calibrated by ensemble simulation: the algebraic  $2 \times 2 \times \delta$  probe was repeated over  $K = 48$  independent codebook-and-value draws on a grid spanning  $c \in [0.08, 1.03]$  at  $D = 64$  and  $D = 128$  (`tests/codebook_ensemble_probe.py`, `tests/varL_calibration.py`). Across the grid, the ensemble standard deviation is well described by  $\text{sd}(L) = \hat{\kappa}(c) A(\delta)/D$ , where  $A(\delta) = 2(1 - \cos \delta)$  is the drift factor of the state/complex law and  $\hat{\kappa}$  ranges from 1.9 to 3.7 over the entire grid, taking values  $\hat{\kappa} \approx 2.4$ – $2.6$  at the operating load  $c \approx 0.30$  for both dimensions and all drift magnitudes. Under this calibration, the trained-model deviations of Table 1 in the main text correspond to approximately one ensemble standard deviation ( $1.02\sigma$  at  $D = 64$ ,  $1.05\sigma$  at  $D = 128$  for the state/complex cell at  $\delta = 0.5$ ): the trained models behave as draws from the same ensemble as the algebraic construction. A complementary eight-codebook probe at  $D = 64$ ,  $n = 20$  (`tests/codebook_ensemble_probe.py`) reports per-codebook maximum absolute deviations between 0.004 and 0.024. The full calibration grid is shipped in `outputs/wave_mem/varL_calibration.json`.

In the key/complex condition, all 30 runs (five seeds  $\times$  six drift magnitudes) produced algebraically zero and numerically null residuals below the numerical guard threshold  $\varepsilon_{\text{null}} = 10^{-4}$ . TOST yielded mean difference  $-0.0031$  with 90% CI  $[-0.0206, 0.0144]$ ; because the lower bound crosses  $-\varepsilon = -0.02$  by 0.0006 under  $n = 5$  seeds, the result is reported as inconclusive with respect to formal equivalence.

## 8 H. Metric audit

A simple mean of per-example ratios  $\frac{1}{N} \sum_i (\|\mathbf{r}_i\|^2 / \|\mathbf{v}_i\|^2)$  introduced large estimation variance in trained models due to small target vector norms in the denominator. The primary metric

was therefore standardized as the pooled ratio of sums  $L = \sum_i \|\mathbf{r}_i\|^2 / \sum_i \|\mathbf{v}_i\|^2$ , stabilizing the estimator.

When the post-erase residual is algebraically zero, computing erased-item exact match by taking an argmax over random noise is uninformative. In our evaluation protocol, residuals below  $\varepsilon_{\text{null}} = 10^{-4}$  are recorded as numerically null residuals; the threshold is set well above the floating-point noise floor of the computation.

## 9 I. Negative results and scope boundaries

The vector-state geometry did not support the intended  $D$ -scaling law. The ideal wave erase and ideal delta erase were algebraically identical under normalized keys. The unnormalized  $\beta = 1$  corrective delta baseline diverged due to unnormalized key norms. Learned codebooks were not included in the closed-form theory. These results define the scope of the present analysis.

## References

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